

SIMATS ENGINEERING

ASSESSMENT TOOL 2

Name: G J D Sai Sharan
COURSE NAME: Deep Learning

Reg no: 192425152
COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour
Total Marks:50

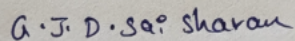
Annexure A Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Code of Conduct:

*I G J D Sai Sharan/192425152 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided

Introduction

A Neural Network is a machine learning model that learns patterns from data to make predictions. For regression problems such as predicting student scores, **Mean Squared Error (MSE)** is commonly used to evaluate model performance.

MSE Formula:

$$\text{MSE} = (1/n) \sum (\text{Actual Value} - \text{Predicted Value})^2$$

A lower MSE indicates better prediction accuracy. Ideally, both the training and testing MSE should be low and close to each other.

Given:

- Training MSE = **0.85**
- Testing MSE = **5.20**

The testing MSE is much higher than the training MSE, indicating that the model does not generalize well to unseen data.

1. Error Identification

The neural network has a very low training error but a high testing error. This indicates that the model performs well on the training dataset but poorly on new data.

Identified Error: Overfitting

Overfitting occurs when the model memorizes the training data instead of learning the underlying patterns, resulting in poor performance on unseen data.

2. Error Analysis and Reasoning

The possible reasons for this difference are:

a) Overfitting

The model has learned the training data too closely, including noise and unnecessary details. Therefore, it cannot make accurate predictions on test data.

b) Complex Model Architecture

A neural network with too many hidden layers or neurons becomes highly complex and is more likely to memorize the training data rather than generalize.

c) Insufficient Training Data

If the dataset contains only a limited number of student records, the network cannot learn enough patterns, leading to poor testing performance.

d) Lack of Regularization

Without techniques such as Dropout or L2 Regularization, the model becomes overly dependent on the training samples and overfits easily.

e) Noisy or Outlier Data

Incorrect marks, missing values, or extreme values in the dataset can increase prediction errors and reduce model accuracy.

f) Improper Data Distribution

If the training and testing datasets have different characteristics, the model may perform well during training but fail during testing.

3. Correction Strategy

The following measures can improve the model:

Apply Regularization

Use L1/L2 Regularization or Dropout to reduce overfitting and improve generalization.

Use Early Stopping

Stop training when the validation error starts increasing to prevent the model from learning noise.

Increase Training Data

Using more student records helps the model learn a wider variety of patterns and improves prediction accuracy.

Simplify the Model

Reduce the number of hidden layers or neurons to decrease unnecessary complexity.

Data Preprocessing

Remove missing values, noisy records, and outliers. Normalize the data before training.

Cross-Validation

Use k-fold cross-validation to evaluate the model more reliably and reduce bias.

4. Interpretation of Results

The training MSE of **0.85** indicates that the model has learned the training data effectively. However, the testing MSE of **5.20** is much higher, showing that the model cannot generalize well to new data.

The large difference between training and testing errors confirms that the neural network is **overfitting**.

After applying regularization, early stopping, simplifying the model, improving the dataset, and using proper validation techniques, the testing MSE is expected to decrease, resulting in better prediction accuracy and improved model performance.

Conclusion

The neural network suffers from **overfitting**, as indicated by the significant difference between the training MSE (0.85) and testing MSE (5.20). The main causes include a complex model, insufficient training data, lack of regularization, noisy data, and differences in data distribution. By applying regularization techniques, early stopping, simplifying the network architecture, improving data quality, and using cross-validation, the model can generalize better, reduce testing error, and achieve more reliable predictions.